

DATA WAREHOUSING & DATA MINING

Quick Revision Pointers · All 4 Units

Unit I: Data
Warehousing & ETL

Unit II: Dimensional
Modeling & OLAP

Unit III: Data Mining &
Pre-Processing

Unit IV: Applications
& Tools

UNIT I — Data Warehousing & ETL Tools

1. What is a Data Warehouse?

- **Inmon's Definition**

Subject-oriented, integrated, time-variant, non-volatile collection of data for decision-making.

- **Subject-Oriented**

Data organised around business subjects (Sales, Customers) — not applications.

- **Integrated**

Data from multiple sources made consistent (unified formats, codes, naming).

- **Time-Variant**

Full history retained — records carry timestamps for trend analysis.

- **Non-Volatile**

Data is only loaded & read — never updated or deleted after loading.

2. OLTP vs. OLAP

Feature	OLTP (Database)	OLAP (Data Warehouse)
Purpose	Day-to-day operations	Analysis & decision-making
Data	Current	Historical
Operations	Insert / Update / Delete	Mostly SELECT (read-only)
Design	Normalised (3NF / ER model)	Star / Snowflake schema
Users	Clerks, data-entry staff	Analysts, managers, executives
Response	Milliseconds	Seconds to minutes

3. Why Data Warehousing? (Key Reasons)

- **Scattered Data**

Data spread across ERP, CRM, inventory — impossible to analyse together.

- **Data Inconsistency**

Different departments use different formats for the same entity.

- **No Historical Analysis**

Operational DBs don't retain long-term history for trend analysis.

- **Single Version of Truth**

DW gives one consistent, authoritative view across the organisation.

- **Performance Issues**

Heavy analytical queries on OLTP slow down day-to-day transactions.

- **Business Intelligence**

Enables trends, forecasts, KPIs, and scorecards for strategic decisions.

4. DW vs. Data Mart

- **Data Warehouse**

Enterprise-wide, terabytes, fully integrated, serves all departments.

- **Data Mart**

Department-specific (e.g. Sales Mart), smaller (GB), quicker to build.

💎 *Think: DW = full superstore; Data Mart = one department inside it.*

5. Three-Tier Architecture

Tier	Name	What it contains
Bottom (1st)	DW Server	Actual warehouse DB + ETL processes
Middle (2nd)	OLAP Server	OLAP engine (ROLAP/MOLAP) — processes queries
Top (3rd)	Front-End / Client	Query tools, reporting tools, dashboards

6. Metadata Types

- **Technical**

Table names, column names, data types, indexes.

- **Business**

Business-friendly names and definitions (e.g. Revenue = Price × Qty).

- **Operational**

ETL job history — when loaded, row counts, errors.

- **Process**

Query performance, data lineage — where did the data come from?

7. ETL — Extract, Transform, Load

- **Extract**

Pull data from source systems (DBs, flat files, APIs). Full or Incremental.

- **Transform — Cleaning**

Fix missing values, remove duplicates, correct errors.

- **Transform — Integration**

Merge data from different sources into one structure.

- **Transform — Aggregation**

Summarise data (e.g. calculate total monthly sales).

- **Transform — Derivation**

Create new fields from existing ones (e.g. Age from DOB).

- **Load — Initial**

First-time full bulk load into an empty warehouse.

- **Load — Incremental**

Periodic loading of only new/changed data — most common.

- **Load — Full Refresh**

Completely replace all data on a schedule.

8. Dimensional Analysis & Information Packages

- **Facts**

Measurable business metrics — Sales Amount, Units Sold, Profit.

- **Dimensions**

Context of facts — Time, Product, Customer, Location.

- **Information Package**

A matrix mapping Dimensions (rows) vs. Facts/Measures (columns) to document DW requirements.

UNIT II — Dimensional Modeling, OLAP & Business Strategy

1. Dimensional Modeling (Ralph Kimball)

- **Goal**

Make data understandable to business users and deliver fast query performance.

- **Step 1**

Choose the business process (e.g. Sales).

- **Step 2**

Choose the grain — level of detail per row (e.g. one row per transaction line).

- **Step 3**

Choose dimensions — who, what, when, where (Customer, Product, Time, Store).

- **Step 4**

Choose facts — the measurements (Revenue, Units Sold, Discount).

2. Schema Types

- **Star Schema ★**

One central Fact Table surrounded by denormalised Dimension Tables. Fastest queries, simple structure.

- **Snowflake Schema ❄️**

Dimension tables normalised into sub-tables. Less storage, more joins, slower queries.

- **Galaxy / Fact Constellation**

Multiple Fact Tables sharing common Dimension Tables. For enterprise multi-process DWs.

Feature	Star Schema	Snowflake Schema
Dimension Tables	Denormalised (flat)	Normalised (split into sub-tables)
Storage	More (redundancy)	Less (no redundancy)
Query Speed	Faster (fewer joins)	Slower (more joins)
Complexity	Simple	More complex

3. OLAP Operations

Operation	Description	Example
Drill-Down	Summary → MORE detail (deeper)	Year → Quarter → Month → Day

Roll-Up	Detail → SUMMARISED (higher)	Day → Month → Quarter → Year
Slice	Fix one dimension value → 2D view	Time=2024; see Product × Region
Dice	Fix ranges on 2+ dims → sub-cube	Time={2023,2024}, Region={Delhi, Mumbai}
Pivot / Rotate	Swap/rotate axes to change perspective	Swap rows (Time) ↔ columns (Product)

♦ *FASMI = Fast Analysis of Shared Multidimensional Information — defines OLAP.*

4. OLAP Models

Feature	MOLAP	ROLAP	DOLAP
Storage	Proprietary multidim DB	Relational DB	Local desktop
Performance	Fastest	Slower	Fast (local)
Scalability	Limited	Highest	Very limited
Best For	Small-medium, pre-computed	Large data, ad-hoc	Individual analysts
Examples	IBM Cognos TM1, Essbase	MicroStrategy	Desktop BI tools

5. Executive Information Systems (EIS)

- **Purpose**

Decision-support system for senior executives — high-level visual performance view.

- **Features**

Highly graphical (charts, gauges, KPI indicators), drill-down capability, no SQL needed.

- **Data Sources**

Accesses DW + external sources (competitor data, economic indicators).

- **Supports**

Critical Success Factors (CSFs) and KPIs monitoring.

6. DW & Business Strategy

- **Single Version of Truth**

Eliminates conflicting reports from different departments.

- **Competitive Advantage**

Identifies profitable customers, products, and markets.

- **CRM**

Understanding customer behaviour and preferences from historical data.

- **Supply Chain**

Analysing supplier performance and inventory patterns.

UNIT III — Data Mining & Data Pre-Processing

1. Data Mining — Definition & Sources

- **Definition**

Non-trivial extraction of implicit, previously unknown, and potentially useful information from large datasets.
(Frawley et al.)

- **Relational DBs**

Tables — most common mining source.

- **Data Warehouses**

Pre-integrated historical data — ideal for mining.

- **Text Data**

Emails, documents, social media — Text Mining.

- **Spatial Data**

Geographic/map data — Spatial Mining.

- **Time-Series**

Stock prices, sensor readings, weather data over time.

- **Web Data**

Web pages, clickstreams, hyperlinks — Web Mining.

2. Patterns That Can Be Mined

Pattern	Description	Use Case
Association Rules	Items frequently occurring together	Retail: bread buyers also buy butter
Classification	Assign data to predefined categories	Medical: tumour benign or malignant?
Clustering	Group similar data, no predefined classes	Marketing: customer segmentation
Regression	Predict a continuous numeric value	Finance: predict house price
Anomaly Detection	Find data points different from norm	Banking: detect fraud
Sequential Patterns	Patterns in ordered/time-stamped data	Camera buyers later buy memory cards

3. Major Issues in Data Mining

- **Efficiency & Scalability**

Algorithms must work on terabyte-scale data in reasonable time.

- **Data Quality**

Incomplete, noisy, inconsistent data → wrong patterns ('Garbage In, Garbage Out').

- **Privacy & Security**

Sensitive data (medical, financial) must not be misused.

- **Data Diversity**

Text, spatial, stream data each need different techniques.

4. Data Pre-Processing Pipeline

- **1. Data Cleaning**

Fix missing values, remove noise and inconsistencies.

- **2. Data Integration**

Combine data from multiple heterogeneous sources.

- **3. Data Reduction**

Reduce data volume while keeping analytical results intact.

- **4. Data Transformation**

Scale, normalise, and aggregate data for mining.

- **5. Data Discretization**

Convert continuous values into discrete categories/intervals.

◆ *Key principle: 'Garbage In, Garbage Out' — pre-processing fixes data quality before mining.*

5. Data Cleaning — Handling Missing Values

- **Ignore the tuple**

Skip entire record — only valid if very few missing values.

- **Global constant**

Fill missing values with 'Unknown' or 0.

- **Mean/Median/Mode**

Replace missing numeric values with the attribute's average or most common value.

- **Most Probable Value**

Use regression or Bayesian inference to predict the missing value.

Noisy Data — Handling Methods

- **Binning**

Sort data, divide into bins; smooth using bin mean, median, or boundaries.

- **Regression**

Fit data to a regression function; outliers pulled towards the line.

- **Clustering**

Find and remove outliers — data points outside all clusters.

6. Data Integration Challenges

- **Schema Integration**

Match equivalent entities across DBs — 'Customer_ID' in DB1 = 'Cust_No' in DB2.

- **Redundancy Detection**

Same attribute stored multiple times — use correlation analysis to detect.

- **Tuple Duplication**

Same record in multiple sources — use deduplication techniques.

7. Data Reduction Techniques

- **Dimensionality Reduction**

Reduce number of attributes — PCA (Principal Component Analysis), feature selection.

- **Numerosity Reduction**

Replace data with smaller representations — regression, histograms, clustering, sampling.

- **Attribute Subset Selection**

Remove irrelevant/redundant attributes — forward selection, backward elimination.

8. Data Transformation

Technique	Description	Formula / Example
Min-Max Normalisation	Scale values to [0,1] range	$(v - \min) / (\max - \min)$
Z-Score Normalisation	Scale based on mean and std deviation	$(v - \text{mean}) / \text{std_dev}$
Decimal Scaling	Divide by power of 10	$v / 10^j$ where $\max(v) < 1$
Generalisation	Replace low-level with higher concepts	Age 23 → 'Young Adult'
Aggregation	Summarise at higher granularity	Daily sales → Monthly sales

9. Data Discretization

- **Equal-Width Binning**

Divide range into N bins of equal width.

- **Equal-Frequency Binning**

Each bin contains the same number of data points.

- **Entropy-Based (Decision Tree)**

Use information gain to find natural cut points — most intelligent method.

- **Cluster Analysis**

Group continuous values into clusters; use cluster labels as categories.

♦ *Example: Age (1–100) → Young (1–25), Middle-Aged (26–55), Senior (56–100)*

10. IRIS Dataset (Standard Benchmark)

- **Samples**

150 total — 50 per class.

- **Classes**

3: Iris Setosa, Iris Versicolor, Iris Virginica.

- **Features**

4: Sepal length, Sepal width, Petal length, Petal width (all in cm).

- **Used For**

Classification, clustering, normalisation, visualisation demos.

11. Apriori Algorithm — Key Concepts

Term	Definition	Formula
Support	How often itemset appears in all transactions	$\text{Txn}(\text{A and B}) / \text{Total Txns}$
Confidence	How often the rule is correct — $P(B A)$	$\text{Txn}(\text{A and B}) / \text{Txn}(\text{A})$
Lift	How much more likely B given A vs. without A	$\text{Confidence}(\text{A} \rightarrow \text{B}) / \text{Support}(\text{B})$
Min Support	User-defined threshold — below this, discard itemset	User-defined
Min Confidence	User-defined threshold — below this, reject rule	User-defined

- **Apriori Property**

All subsets of a frequent itemset must also be frequent. If an itemset is infrequent, ALL its supersets are infrequent
→ enables pruning.

- **Step 1**

Scan DB, count support of all single items → keep frequent 1-itemsets (L1).

- **Step 2**

Generate candidate 2-itemsets (C2) by joining L1 with itself.

- **Step 3**

Scan DB, count support of C2, discard below min_sup → get L2.

- **Step 4**

Repeat: C(k+1) from L(k), count, prune → L(k+1).

- **Step 5**

Stop when no new frequent itemsets found.

- **Step 6**

Generate association rules from frequent itemsets where confidence $\geq$ min_conf.

♦ *Example: min_sup=50%, min_conf=60%. Rule: Bread → Milk with Confidence=66% → valid rule.*

UNIT IV — Data Mining Applications & Tools

1. Data Mining in Retail

Application	Description	Algorithm
Market Basket Analysis	Products frequently bought together	Apriori, FP-Growth
Customer Segmentation	Group customers by behaviour/demographics	K-Means Clustering
Churn Prediction	Identify customers likely to stop buying	Decision Trees, Logistic Regression
Recommender Systems	'Customers who bought X also bought Y'	Collaborative Filtering
Price Optimisation	Analyse how price changes affect sales	Regression Analysis

2. Data Mining in Telecom

- **Churn Analysis**

Predict which subscribers will switch — trigger proactive retention offers.

- **Network Fault Prediction**

Analyse patterns to predict equipment failures — reduce downtime.

- **Fraud Detection**

Identify unusual calling patterns — clone fraud, international fraud.

- **Usage Pattern Analysis**

Analyse call patterns to design better tariff plans.

3. Recommender Systems

Type	How it works	Limitation
Collaborative Filtering	Based on what similar users liked	Cold start: new users/items have no history
Content-Based Filtering	Based on attributes of items you liked	Recommends only similar items — no diversity
Hybrid Systems	Combines both approaches	More complex to implement

♦ *Netflix, Amazon, Spotify all use Hybrid Recommender Systems.*

4. WEKA (Waikato Environment for Knowledge Analysis)

- **Developer / License**

University of Waikato, New Zealand — Free, Open Source (GNU GPL), Java.

- **File Format**

ARFF (Attribute-Relation File Format) and CSV.

- **Classification Algos**

J48 (Decision Tree/C4.5), Naïve Bayes, SVM (SMO), kNN (IBk), Random Forest.

- **Clustering Algos**

k-Means, EM (Expectation Maximisation), DBSCAN, Cobweb.

- **Association Rules**

Apriori, FP-Growth.

IRIS in WEKA (Exam Steps):

- **Step 1**

Open WEKA Explorer → load iris.arff.

- **Step 2**

Classify tab → trees → J48.

- **Step 3**

Test options → Cross-validation, Folds=10.

- **Step 4**

Click Start → view accuracy, confusion matrix, precision, recall, F-measure.

- **Step 5**

Right-click result → visualise the decision tree.

5. RapidMiner

- **Developer / License**

RapidMiner Inc. — Community edition free, Enterprise paid, Java.

- **Interface**

Visual drag-and-drop workflow — no coding required.

- **Process**

A complete workflow of connected operators from input to output.

- **Operator**

A building block: 'Read CSV', 'Normalise', 'Decision Tree', 'Write Excel'.

- **Port**

Input/output connectors that link operators in the pipeline.

IRIS in RapidMiner (Exam Steps):

- **Step 1**

Import IRIS using Read CSV operator.

- **Step 2**

Set Role operator → set 'class' as label (target variable).

- **Step 3**

Decision Tree inside a Cross Validation block.

- **Step 4**

Connect to Performance operator to measure accuracy.

- **Step 5**

Run → view accuracy, confusion matrix, ROC curve.

♦ *RapidMiner vs WEKA: More intuitive visual interface, better for large datasets, more enterprise integration options.*

6. IBM Watson (Enterprise AI/ML Platform)

- **Type**

Cloud-based — Watson Studio + Watson Machine Learning.

- **Key Feature**

AutoAI — automatically builds, trains, selects, and deploys best ML model.

- **Integrations**

IBM Cloud, Apache Spark, scikit-learn, TensorFlow, Keras.

- **Deployment**

Deploy models as REST APIs callable from any application.

- **Lifecycle**

Full ML: data prep → model building → deployment → monitoring.

IRIS Classification in Watson (Exam Steps):

- **Step 1**

Upload IRIS dataset to Watson Studio as a data asset.

- **Step 2**

Create AutoAI Experiment → select 'class' as target column.

- **Step 3**

Watson tries: Random Forest, XGBoost, Decision Tree, SVM, Logistic Regression.

- **Step 4**

Watson ranks models by accuracy → selects best pipeline.

- **Step 5**

Save and Deploy → exposes model as REST API endpoint.

- **Step 6**

Call API with new iris measurements to classify species.

7. Classification & Clustering Algorithms

- **Decision Tree (J48/C4.5)**

Recursively splits data using Information Gain (entropy reduction). Produces human-readable if-then rules. Prone to overfitting — use pruning.

- **k-Nearest Neighbours (kNN)**

Classify new point by majority class of k nearest neighbours (Euclidean distance). Lazy learner — no training phase, computes at query time.

- **Naïve Bayes**

Probabilistic classifier using Bayes' Theorem. Assumes all features independent. Works very well for spam detection, sentiment analysis, text classification.

- **k-Means Clustering**

Init k centroids → assign each point to nearest → recalculate centroids → repeat until convergence. With IRIS k=3 closely matches 3 actual species.

Quick Comparison — Tools at a Glance

Feature	WEKA	RapidMiner	IBM Watson
Interface	GUI + command line	Visual drag-and-drop	Cloud web UI
Ease of Use	Moderate	Easy (no coding)	Very easy (AutoAI)
Best For	Academic research	Business analytics	Enterprise ML
Deployment	Local only	Local + server	Cloud REST API
Automation	Manual	Semi-automated	Fully automated (AutoAI)
Cost	Free (open source)	Free + paid tiers	Paid (limited free)
IRIS Demo	J48 Classify → Eval	Decision Tree in CV	AutoAI auto-selects best

10 Must-Know Definitions

- **Data Warehouse**

Subject-oriented, integrated, time-variant, non-volatile collection of data for management decision-making (Inmon).

- **ETL**

Extract, Transform, Load — move data from source systems, clean/reformat, and load into the warehouse.

- **OLAP**

Online Analytical Processing — interactive multidimensional analysis enabling complex calculations and trend analysis.

- **Data Mining**

Non-trivial extraction of implicit, previously unknown, and potentially useful information from large datasets.

- **Association Rule**

A rule $A \rightarrow B$ meaning 'if A then B' with quantified Support and Confidence.

- **Apriori Property**

All subsets of a frequent itemset must be frequent; if an itemset is infrequent, all supersets are also infrequent.

- **Star Schema**

DW schema with one central Fact Table surrounded by denormalised Dimension Tables.

- **MOLAP**

Multidimensional OLAP — data in proprietary multidimensional DB; fastest queries but limited scalability.

- **ROLAP**

Relational OLAP — OLAP engine on relational DB; most scalable, handles very large data volumes.

- **Normalisation**

Scaling attribute values to a common range (e.g. [0,1]) to eliminate scale bias in distance-based mining.